# Practice workbook · Sound sampling: analogue to digital

How analogue sound becomes digital: the sampling process, reading/plotting a sampled waveform and its axes, sampling frequency and the effect of changing it.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-13

# 1. Question 1 · part c.i · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

**1** Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

**(2)**

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

**(2)**

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(iii) Convert the binary pattern 11000110 to hexadecimal.

**(2)**

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# 1. Question 1 · part c.i · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

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(b) Images are stored as binary data.

(i) A bitmap image is 400 pixels wide by 200 pixels high.

It has a colour depth of 12 bits.

Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

**(2)**

(ii) **Figure 1** is an image of an icon, set out in an 8 × 8 pixel grid.

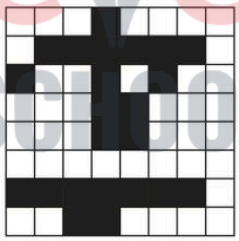


**Figure 1**

A run-length encoding compression algorithm uses:

- W for white pixels
- B for black pixels.

Give the result of compressing the first 4 lines of the icon using the algorithm.

**(3)**

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

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# 1. Question 1 · part c.i · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

(iii) Run-length encoding is a type of lossless compression.

State what is meant by the term **lossless** in this context.

(1)

(c) Raoul wants to record a song and store it as a digital sound file.

**Figure 2** shows an overview of the process of recording a song and storing it as a digital sound file.

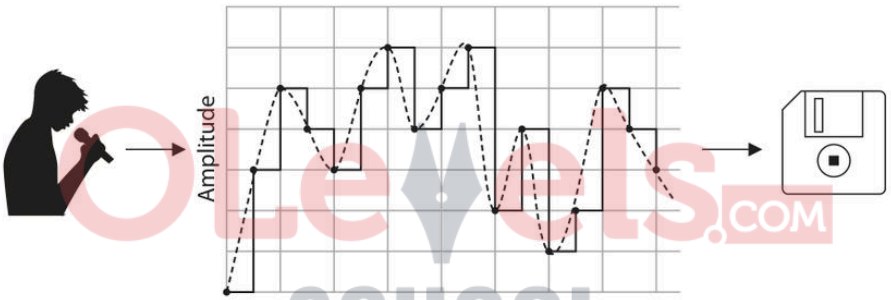



**Figure 2**

(i) Describe the steps taken to convert the analogue sound to a digital sound file.

(3)

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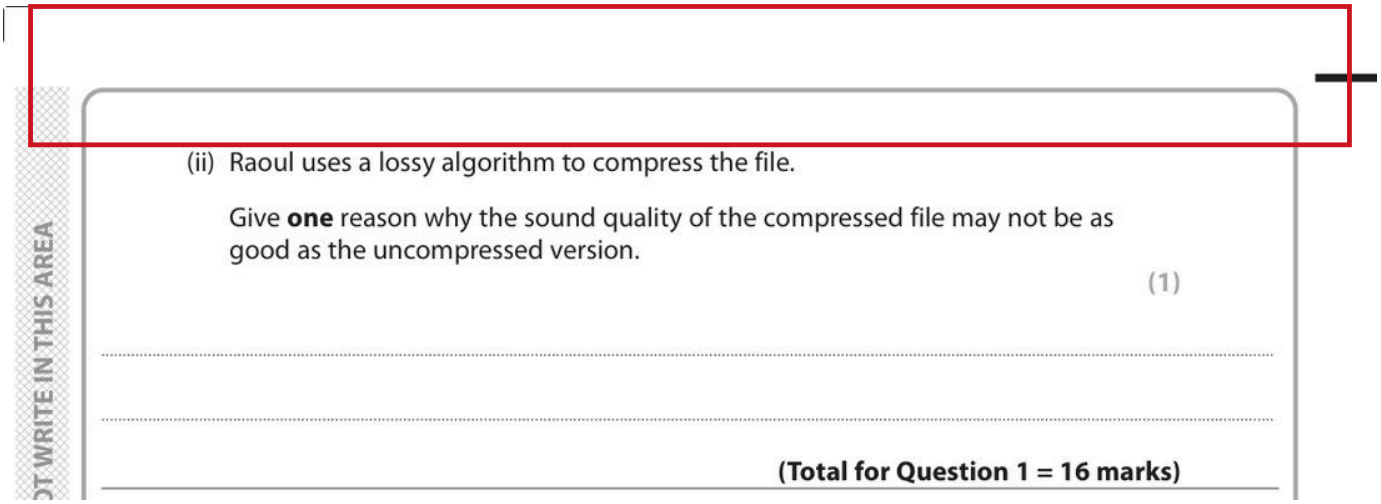


P 6 9 2 9 7 A 0 4 2 0

# 1. Question 1 · part c.i · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.



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(ii) Raoul uses a lossy algorithm to compress the file.

Give **one** reason why the sound quality of the compressed file may not be as good as the uncompressed version.

**(1)**

.....

.....

**(Total for Question 1 = 16 marks)**

Working and answer

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## 2. Question 4 · part a.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.i only.

**4** Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

**Figure 3** illustrates this process.

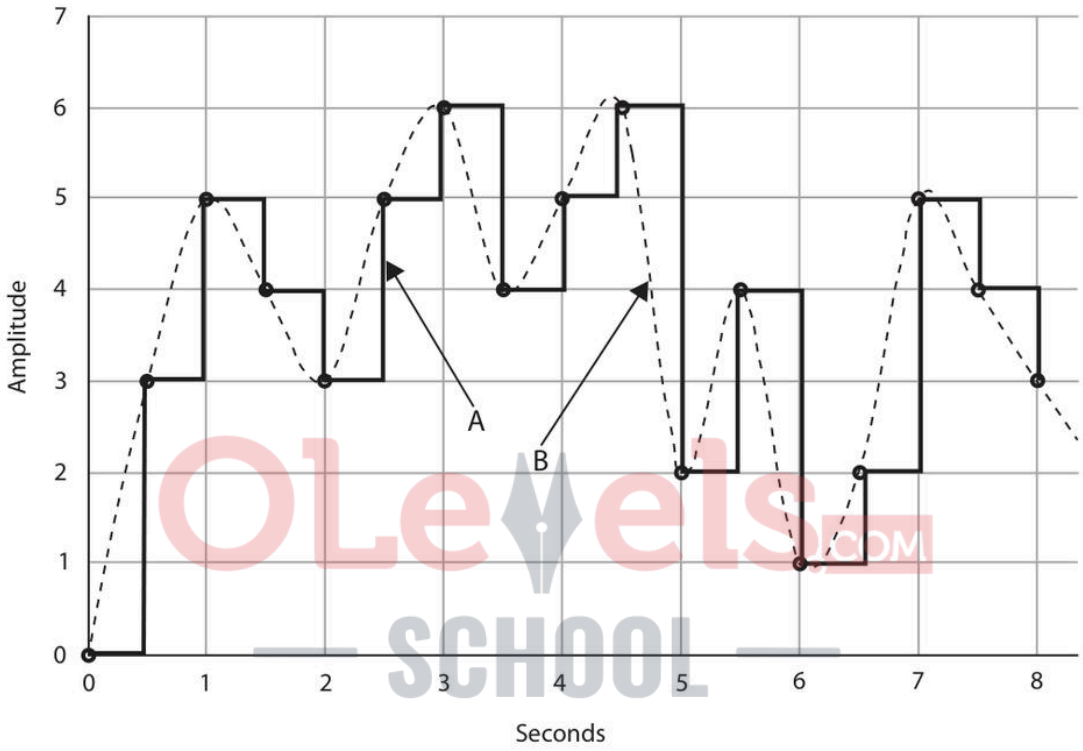



**Figure 3**

(i) State what the solid line labelled **A** represents.

**(1)**

(ii) State what the dashed line labelled **B** represents.

**(1)**

(iii) Give the sampling frequency, including the correct units.

**(1)**

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P 6 1 8 8 4 R R A 0 1 0 2 0

## 2. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

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## 3. Question 4 · part a.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.ii only.

**4** Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

**Figure 3** illustrates this process.

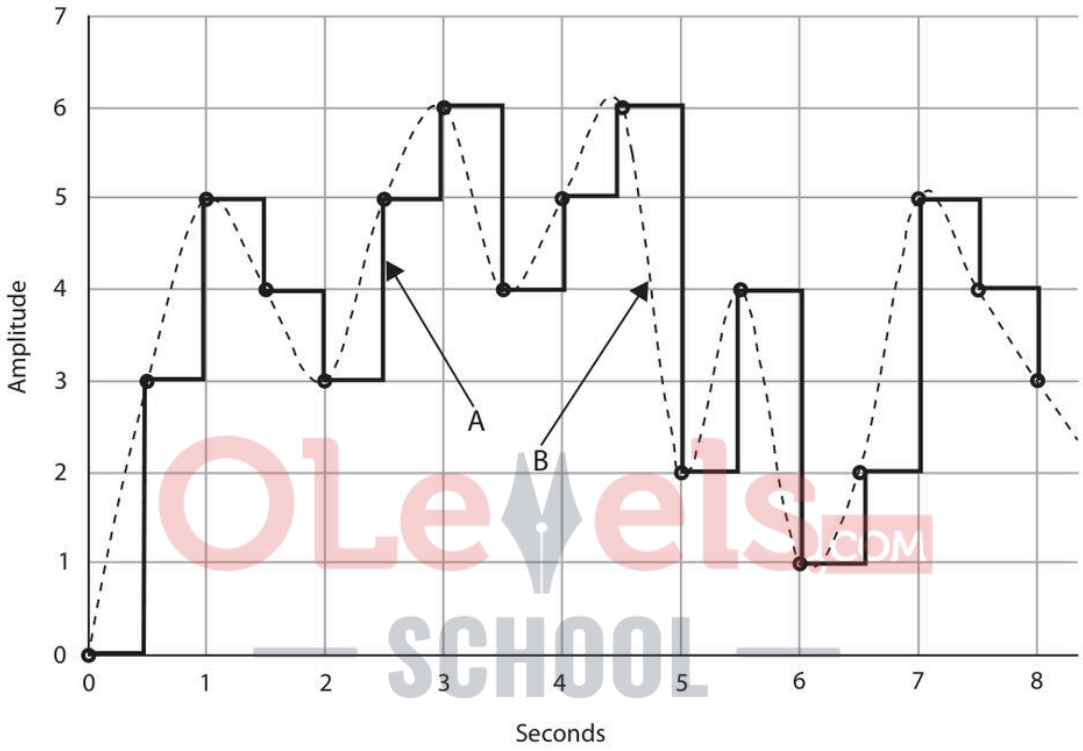



**Figure 3**

(i) State what the solid line labelled **A** represents.

**(1)**

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**(1)**

(iii) Give the sampling frequency, including the correct units.

**(1)**

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P 6 1 8 8 4 R R A 0 1 0 2 0

## 3. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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# 4. Question 4 · part a.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.iii only.

**4** Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

**Figure 3** illustrates this process.

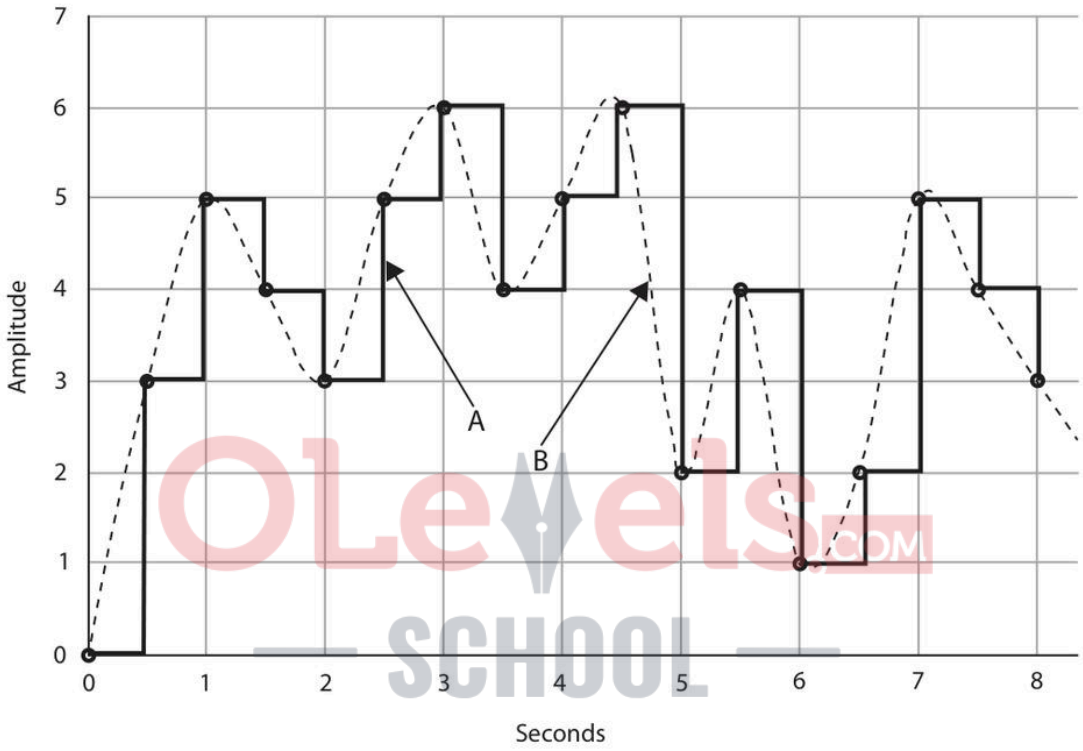



**Figure 3**

(i) State what the solid line labelled **A** represents.

**(1)**

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**(1)**

(iii) Give the sampling frequency, including the correct units.

**(1)**

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P 6 1 8 8 4 R R A 0 1 0 2 0

# 4. Question 4 · part a.iii · 1 mark

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Further practice · Attempt part a.iii only.

(iv) Give **one** impact of increasing the sampling frequency.

**(1)**

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

**(2)**

Bit-depth

Justification

(vi) Give the value in binary of the audio sample at the 4th second.

**(1)**

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## 4. Question 4 · part a.iii · 1 mark · Working and answer

Working and answer

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# 5. Question 4 · part a.iv · 1 mark

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Further practice · Attempt part a.iv only.

**4** Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

**Figure 3** illustrates this process.

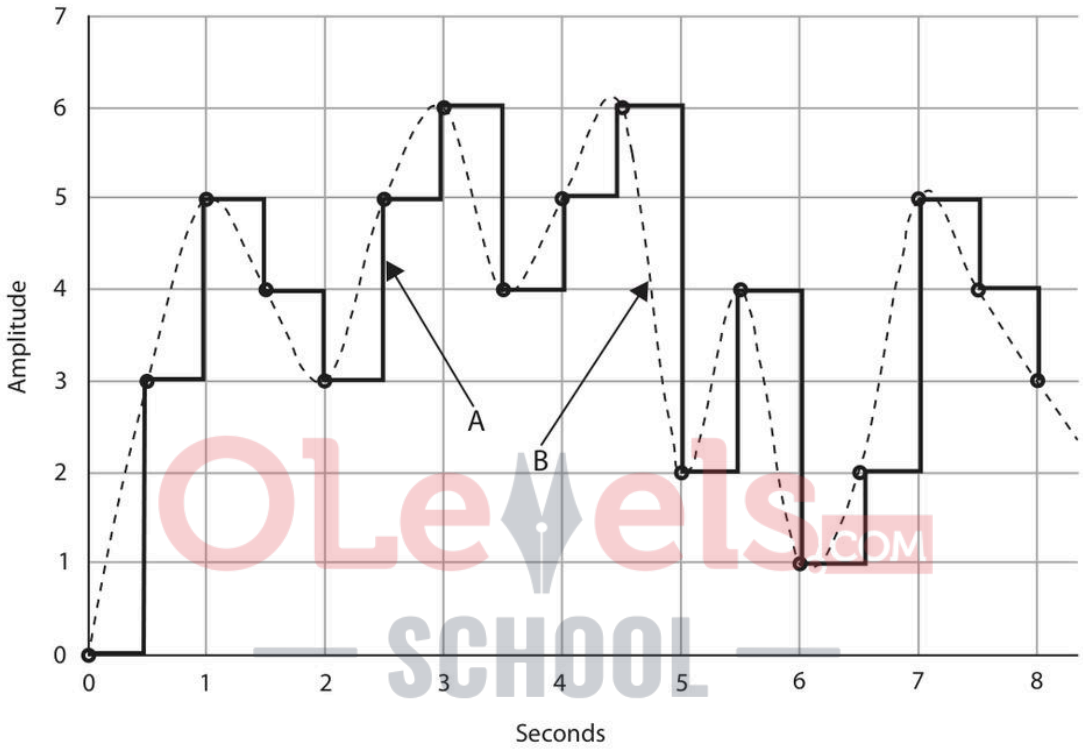



**Figure 3**

(i) State what the solid line labelled **A** represents.

**(1)**

(ii) State what the dashed line labelled **B** represents.

**(1)**

(iii) Give the sampling frequency, including the correct units.

**(1)**

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## 5. Question 4 · part a.iv · 1 mark

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Further practice · Attempt part a.iv only.

(iv) Give **one** impact of increasing the sampling frequency.

(1)

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

(2)

Bit-depth

Justification

(vi) Give the value in binary of the audio sample at the 4th second.

(1)

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## 5. Question 4 · part a.iv · 1 mark · Working and answer

Working and answer

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# 6. Question 3 · part a.i · 3 marks

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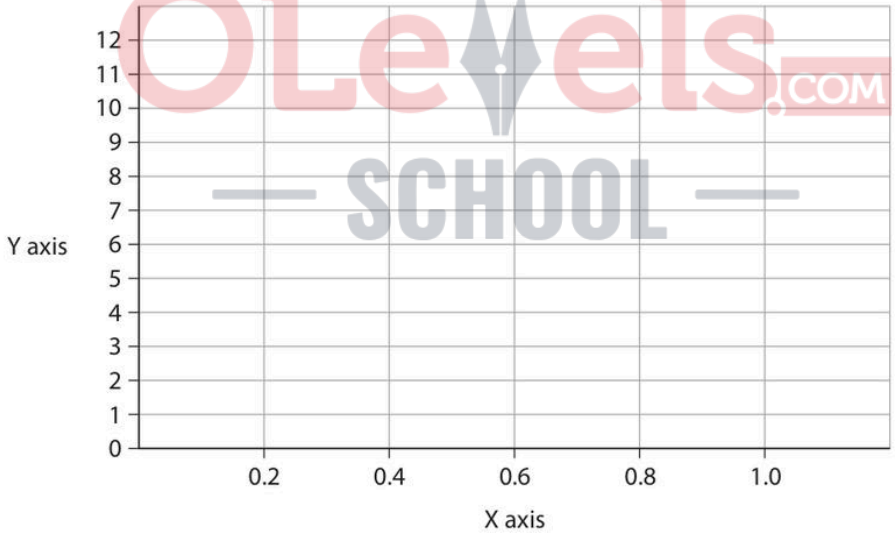
Further practice · Attempt part a.i only.

**3** Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

| Sample number | Denary value |
|---|---|
| 1 | 1 |
| 2 | 10 |
| 3 | 12 |
| 4 | 5 |
| 5 | 3 |

**Figure 1**




(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

**(3)**

(ii) Give a suitable label for the X axis.

**(1)**

(iii) Give a suitable label for the Y axis.

**(1)**

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## 6. Question 3 · part a.i · 3 marks · Working and answer

Working and answer

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## 7. Question 3 · part a.ii · 1 mark

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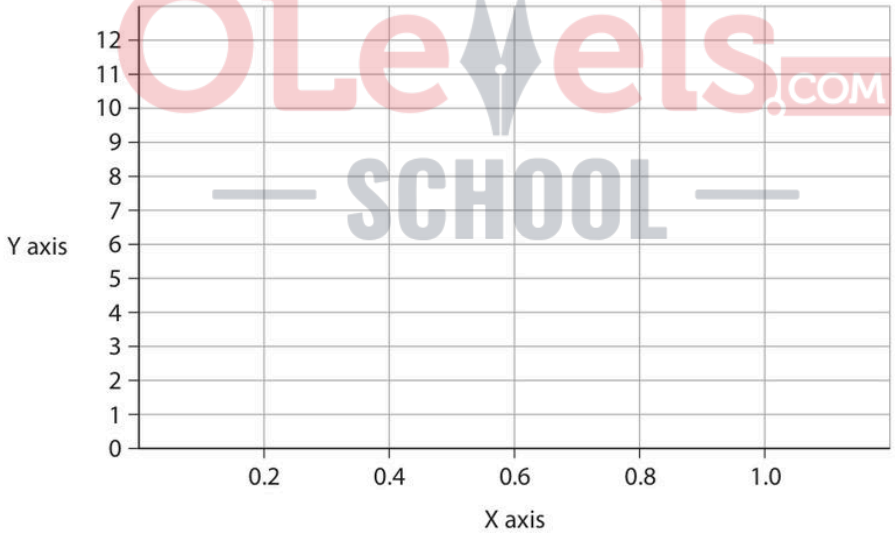
Further practice · Attempt part a.ii only.

**3** Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

| Sample number | Denary value |
|---|---|
| 1 | 1 |
| 2 | 10 |
| 3 | 12 |
| 4 | 5 |
| 5 | 3 |

**Figure 1**




(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

**(3)**

(ii) Give a suitable label for the X axis.

**(1)**

.....

(iii) Give a suitable label for the Y axis.

**(1)**

.....

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P 7 2 5 3 8 A 0 8 2 0

## 7. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

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# 8. Question 3 · part a.iii · 1 mark

2022 JUN · Paper 01

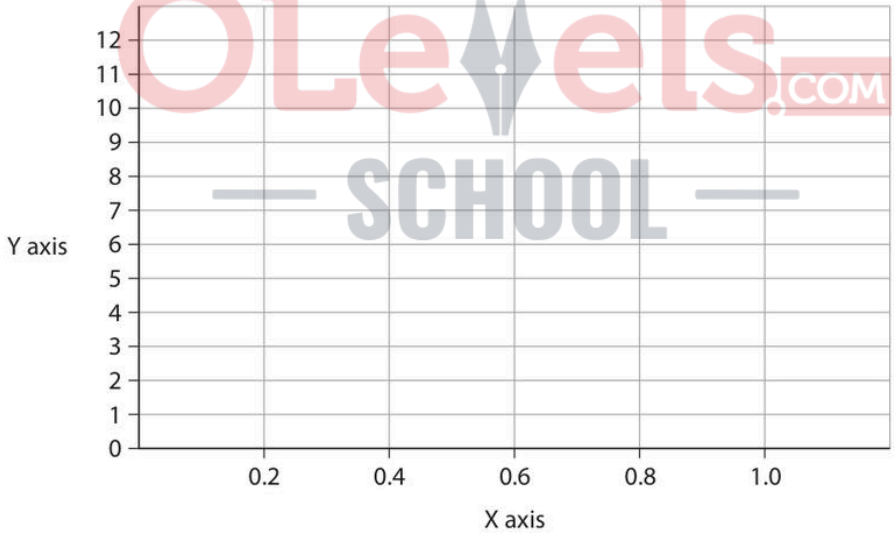
Further practice · Attempt part a.iii only.

**3** Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

| Sample number | Denary value |
|---|---|
| 1 | 1 |
| 2 | 10 |
| 3 | 12 |
| 4 | 5 |
| 5 | 3 |

**Figure 1**




(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

**(3)**

(ii) Give a suitable label for the X axis.

**(1)**

(iii) Give a suitable label for the Y axis.

**(1)**

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P 7 2 5 3 8 A 0 8 2 0

## 8. Question 3 · part a.iii · 1 mark

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Further practice · Attempt part a.iii only.

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(b) Alyssa uploads music files to her cloud storage.

(i) She compresses the files before she uploads them using a lossless algorithm.

Give **one** disadvantage of using a lossless rather than a lossy algorithm for this purpose.

(1)

.....

.....

(ii) Explain **one** benefit to Alyssa of storing her music files in the cloud.

(2)

.....

.....

.....

.....

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(iii) Give **one** possible security issue associated with storing music files in the cloud.

(1)

.....

.....

(iv) One of Alyssa's music files is stored at https://www.cloudisfab.com/re12/ru2.mp3

Complete the table by adding a description of each URL component.

(4)

| URL component | Description |
|---|---|
| https | |
| www.cloudisfab.com | |
| re12 | |
| ru2.mp3 | |

**(Total for Question 3 = 13 marks)**

## 8. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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